

```
import numpy as np
```

```
#view vs copy  
# view means modification in original data while copy means  
# changes in duplicate data will not reflect original data
```

```
a=np.array([10,20,30,40,50,60,70,80,90])  
a
```

```
array([10, 20, 30, 40, 50, 60, 70, 80, 90])
```

```
b=a[3:6]  
b[:]=0
```

```
a
```

```
array([10, 20, 30, 40, 50, 60, 70, 80, 90])
```

```
#copy  
a=np.array([10,20,30,40,50,60,70,80,90])  
a
```

```
array([10, 20, 30, 40, 50, 60, 70, 80, 90])
```

```
b=a[3:6].copy()  
b[:]=0
```

```
a
```

```
array([10, 20, 30, 40, 50, 60, 70, 80, 90])
```

```
#np.arange(min,max,step)==>
```

```
a=np.arange(1,10,3)  
a
```

```
array([1, 4, 7])
```

```
a=np.arange(1,51,2)  
a
```

```
array([ 1,  3,  5,  7,  9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33,  
       35, 37, 39, 41, 43, 45, 47, 49])
```

```
a=np.arange(1,5).reshape(2,2)  
a
```

```
array([[1, 2],  
       [3, 4]])
```

```
b=np.arange(5,9).reshape(2,2)  
b
```

```
array([[5, 6],  
       [7, 8]])
```

```
a**2
```

```
array([[ 1,  4,  9],  
       [16, 25, 36],  
       [49, 64, 81]])
```

```
a**3
```

```
array([[ 1,  8, 27],  
       [ 64, 125, 216],  
       [343, 512, 729]])
```

```
a=np.arange(1,10).reshape(3,3)  
a
```

```
array([[1, 2, 3],
       [4, 5, 6],
       [7, 8, 9]])
```

```
print('total sum of matrix',np.sum(a))
print('row wise sum',np.sum(a,axis=1))
print('col wise sum',np.sum(a,axis=0))
```

```
total sum of matrix 45
row wise sum [ 6 15 24]
col wise sum [12 15 18]
```

```
#unique()==>it will remove duplicate elements from an array
#a==>unique elemnt array
#return index=true==>return indexing of unique ele.
#return count=true==>total frequency of each uniq item
```

```
a=np.array([1,2,3,1,1,1,2,4,4,4,4,4,4])
a
```

```
array([1, 2, 3, 1, 1, 1, 2, 4, 4, 4, 4, 4, 4])
```

```
np.unique(a,return_index=True,return_counts=True)
```

```
(array([1, 2, 3, 4]), array([0, 1, 2, 7]), array([4, 2, 1, 6]))
```

```
#horizontal and vertical scaling
#hor.scaling mean combined 2 0r more arrays in horizontaly
#and ver. scal. mean combined 2 or more aarrays in ver.
```

```
a=np.arange(1,5)
a
```

```
array([1, 2, 3, 4])
```

```
b=np.arange(5,9)
c=np.arange(9,13)
```

```
np.hstack((a,b,c))
```

```
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12])
```

```
np.vstack((a,b,c))
```

```
array([[ 1,  2,  3,  4],
       [ 5,  6,  7,  8],
       [ 9, 10, 11, 12]])
```

```
#matrix (11-50),elements==>matrix 4*4==> rsum,csum,total sum
#mtrix(11-20) ,mtrix(31-40)==>reshape(2*5)==>a+b
```

```
a=np.array([[11,12,13,14],
            [15,17,19,20],
            [22,23,24,25],
            [30,31,32,33]])
a
```

```
array([[11, 12, 13, 14],
       [15, 17, 19, 20],
       [22, 23, 24, 25],
       [30, 31, 32, 33]])
```

```
print('rsum',np.sum(a,axis=1))
print('cs',np.sum(a,axis=0))
print(np.sum(a))
```

```
rsum [ 50  71  94 126]
cs [78 83 88 92]
341
```

```
a=np.arange(11,21).reshape(5,2)
a
```

```
array([[11, 12],
       [13, 14],
       [15, 16],
```

```
[17, 18],  
[19, 20]])
```

```
b=np.arange(31,41).reshape(5,2)  
b
```

```
array([[31, 32],  
       [33, 34],  
       [35, 36],  
       [37, 38],  
       [39, 40]])
```

```
a+b
```

```
array([[42, 44],  
       [46, 48],  
       [50, 52],  
       [54, 56],  
       [58, 60]])
```

```
a-b
```

```
array([[-20, -20],  
       [-20, -20],  
       [-20, -20],  
       [-20, -20],  
       [-20, -20]])
```

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